

DATA301-DATA471: Data Science in Practice

Assignment 4

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1 Exercise 1

We have i.i.d. samples $\mathbf{X} = (X_1, \dots, X_n)$ from an Exponential distribution with rate $\lambda > 0$ (so λ is the reciprocal of the mean). We compare three estimators of λ : the maximum likelihood estimator $\hat{\lambda}_{ML}$, the inter-quartile-range estimator $\hat{\lambda}_{IQR}$, and a bootstrap bias-corrected version of the latter, $\tilde{\lambda}$.

Note. The assignment PDF prints $\hat{\lambda}_{ML}(\mathbf{X}) = \frac{1}{n} \sum_i X_i$, which is just the sample mean $\bar{X}$ (an estimator of $1/\lambda$, not of λ). Since the question statement itself defines λ as “the reciprocal of the mean,” and the true MLE of the rate of an Exponential distribution is $\hat{\lambda}_{ML} = 1/\bar{X}$, we treat this as a typo in the brief and use $\hat{\lambda}_{ML}(\mathbf{X}) = 1/\bar{X} = n/\sum_i X_i$ throughout.

1.1 Order statistics for the quartiles [10%]

TODO: summary of Wicklin (2017) and Hyndman and Fan (1996) — discrete (type 1-3) vs interpolation-based (type 4-9) definitions, H&F’s recommendation of type 8, and the default types used by R (`quantile()`, type 7) and Python (`numpy.percentile, method="linear"`, equivalent to type 7).

```
quantile(1:10) # uses R's default type
```

0%	25%	50%	75%	100%
1.00	3.25	5.50	7.75	10.00

```
identical(quantile(1:10), quantile(1:10, type = 7)) # confirms default = type 7
```

```
[1] TRUE
```

1.2 Bootstrap bias-corrected estimator [20%]

Derivation. For a fixed sample $\mathbf{X}$ of size n , draw B bootstrap resamples $\mathbf{X}_1^*, \dots, \mathbf{X}_B^*$ (sampling n values from $\mathbf{X}$ with replacement). The bootstrap estimate of the bias of $\hat{\lambda}_{IQR}$ is

$$\widehat{\text{bias}} = \frac{1}{B} \sum_{b=1}^B \hat{\lambda}_{IQR}(\mathbf{X}_b^*) - \hat{\lambda}_{IQR}(\mathbf{X}).$$

The first-order bias-corrected estimator subtracts this estimated bias:

$$\tilde{\lambda}(\mathbf{X}) = \hat{\lambda}_{IQR}(\mathbf{X}) - \widehat{\text{bias}} = 2\hat{\lambda}_{IQR}(\mathbf{X}) - \frac{1}{B} \sum_{b=1}^B \hat{\lambda}_{IQR}(\mathbf{X}_b^*).$$

```

lambda_ml <- function(x) 1 / mean(x)

lambda_iqr <- function(x, type = 8) {
  q <- quantile(x, probs = c(0.25, 0.75), type = type)
  log(3) / (q[[2]] - q[[1]])
}

lambda_tilde <- function(x, B = 200, type = 8) {
  n <- length(x)
  boot_est <- replicate(B, lambda_iqr(sample(x, n, replace = TRUE), type = type))
  # For very small n, a bootstrap resample can tie at the quartiles (q3 = q1),
  # making lambda_iqr(.) = Inf; we drop these degenerate resamples from the
  # bootstrap average rather than let a single Inf dominate the correction.
  boot_est <- boot_est[is.finite(boot_est)]
  2 * lambda_iqr(x, type = type) - mean(boot_est)
}

x <- rexp(30, rate = 2)
c(ML = lambda_ml(x), IQR = lambda_iqr(x), bias_corrected = lambda_tilde(x))

```

ML	IQR	bias_corrected
1.562896	1.741311	1.814164

1.3 Monte Carlo comparison [20%]

TODO: justify the number of Monte Carlo replications per (λ, n) cell using the CLT-based precision argument in Robert and Casella (2009) (Ch. 3): the standard error of a Monte Carlo estimate of MSE is $\sqrt{\widehat{\text{Var}}/R}$, so R is chosen so this is small relative to the MSE values being compared. Smaller n and more extreme λ give higher-variance estimators and need larger R .

```

mc_mse <- function(lambda, n, R, B = 200, type = 8) {
  est <- matrix(NA_real_, nrow = R, ncol = 3,
               dimnames = list(NULL, c("ML", "IQR", "bias_corrected")))
  for (r in seq_len(R)) {
    x <- rexp(n, rate = lambda)
    est[r, ] <- c(lambda_ml(x), lambda_iqr(x, type = type), lambda_tilde(x, B = B, type = type))
  }
  colMeans((est - lambda)^2)
}

lambda_grid <- c(0.1, 1, 10)
n_grid <- c(5, 30, 100)
R_grid <- c(`5` = 3000, `30` = 1500, `100` = 800)

results <- do.call(rbind, lapply(lambda_grid, function(lam) {
  do.call(rbind, lapply(n_grid, function(n) {
    mse <- mc_mse(lam, n, R = R_grid[[as.character(n)]])
    data.frame(lambda = lam, n = n, R = R_grid[[as.character(n)]],
               t(mse), check.names = FALSE)
  }))
}))
results

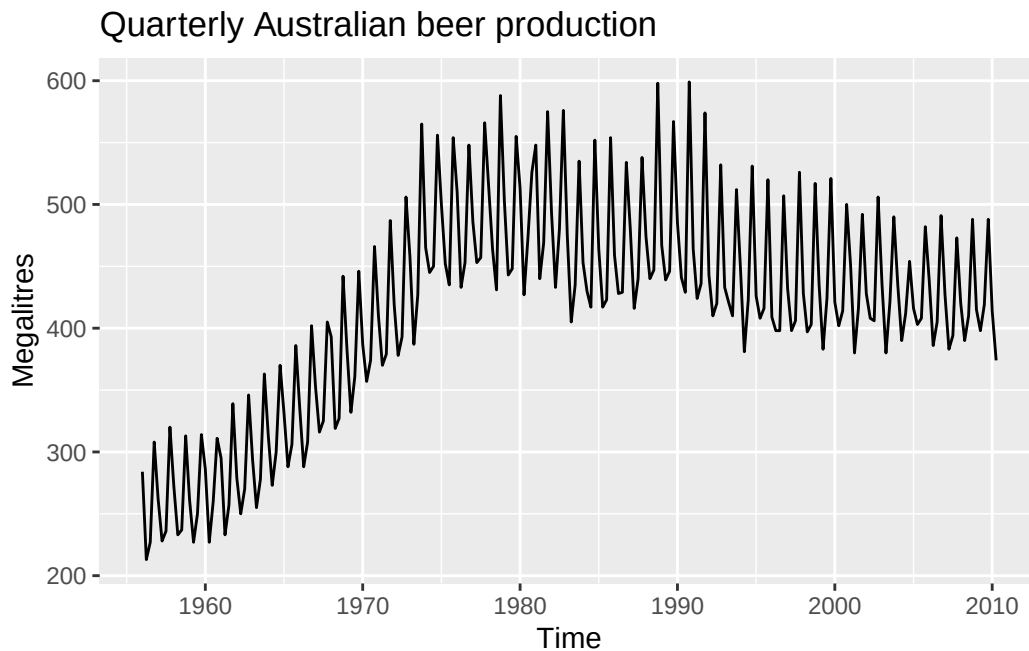
```

	lambda	n	R		ML	IQR	bias_corrected
1	0.1	5	3000	7.120336e-03	1.351785e-02	2.865356e+01	
2	0.1	30	1500	4.041677e-04	9.086582e-04	1.139732e-03	
3	0.1	100	800	1.130194e-04	2.479827e-04	2.911600e-04	
4	1.0	5	3000	6.264403e-01	1.022698e+00	2.951111e+03	
5	1.0	30	1500	3.948991e-02	8.950896e-02	1.090623e-01	
6	1.0	100	800	1.064529e-02	2.492353e-02	2.961271e-02	
7	10.0	5	3000	6.113751e+01	9.969845e+01	1.032500e+05	
8	10.0	30	1500	3.773153e+00	9.334983e+00	1.167701e+01	
9	10.0	100	800	1.154352e+00	2.209206e+00	2.600836e+00	

TODO: discuss the table above (which estimator wins in each corner of the design, whether the bias correction actually helps $\hat{\lambda}_{IQR}$, small- n vs large- n behaviour).

2 Exercise 2

```
autoplot(ausbeer) + labs(title = "Quarterly Australian beer production", y = "Megalitres")
```



2.1 Frequency

TODO

2.2 Trend

TODO

2.3 Seasonality

TODO

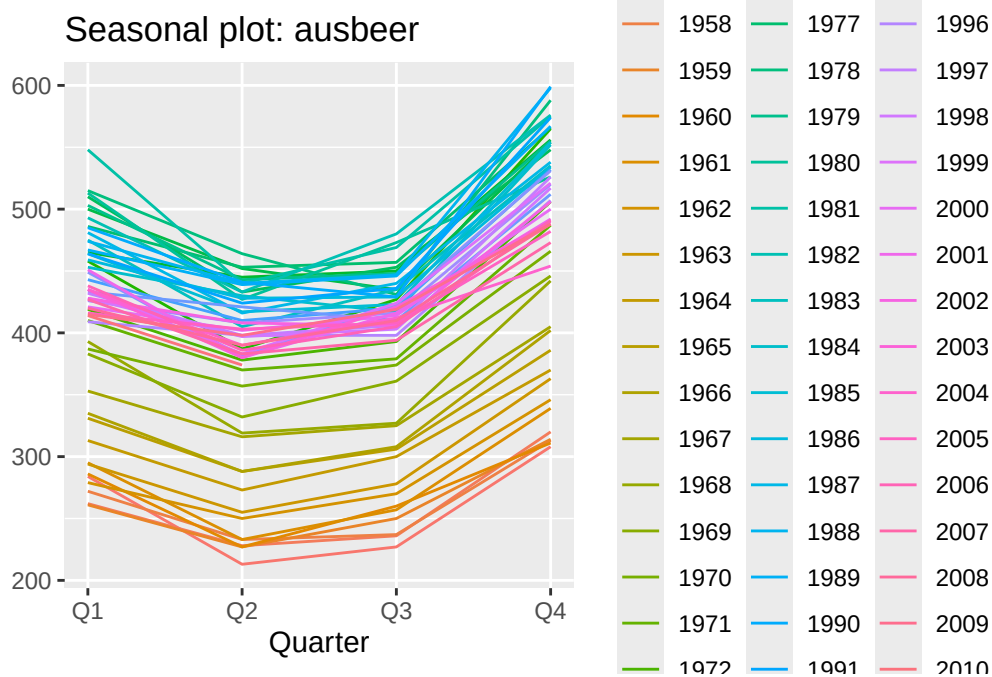
2.4 Variance stability

TODO

2.5 Unusual observations

TODO

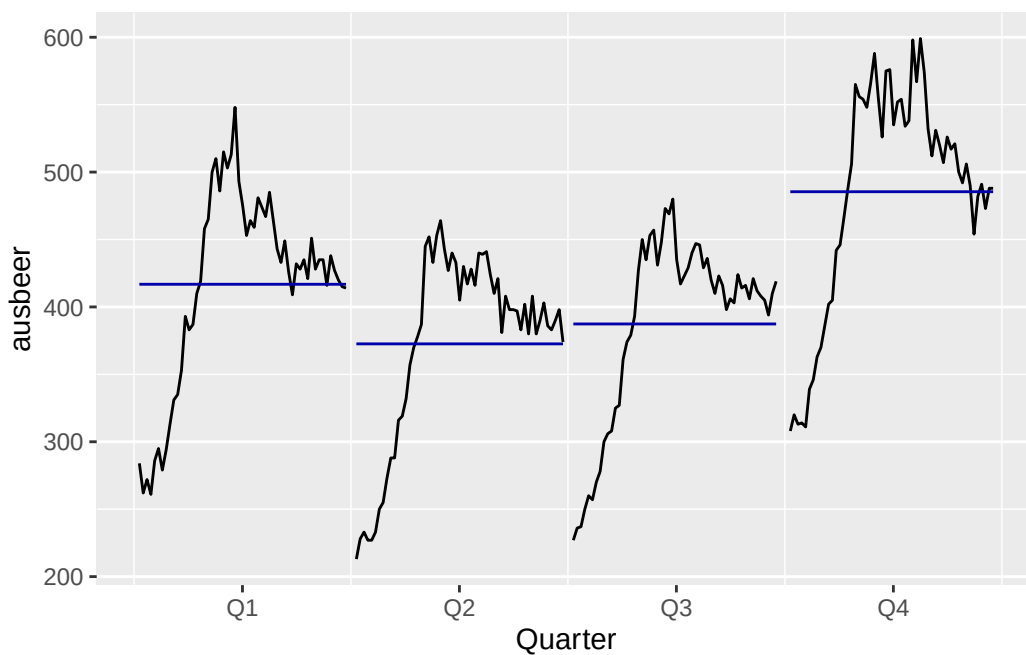
```
ggseasonplot(ausbeer)
```



2.6 Seasonal plot

TODO

```
ggsubseriesplot(ausbeer)
```



2.7 Seasonal subseries plot

TODO

2.8 Detrend / transform?

TODO

References

- Hyndman, Rob J., and Yanan Fan. 1996. “Sample Quantiles in Statistical Packages.” *The American Statistician* 50 (4): 361–65. <https://doi.org/10.2307/2684934>.
- Robert, Christian P., and George Casella. 2009. *Introducing Monte Carlo Methods with R*. Springer.
- Wicklin, Rick. 2017. *Sample Quantiles: A Comparison of 9 Definitions*. <https://blogs.sas.com/content/iml/2017/05/24/definitions-sample-quantiles.html>.